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Printing solutions offer too much of a good thing—with options galore, the trick lies in choosing the best one for your business. The entry of colour laser printers has further enriched choice. Is the marriage of colour with the speed of a laser solution a fruitful one? We try to find out. To make things even more interesting, we also test some inkjet printers meant for business use.

The Printers

Printing solutions have evolved a great deal from the dot-matrix technology of past. A dot-matrix is a low-cost, low-maintenance solution for sure, but due to noisy operation and sub-par print quality, it is rarely seen outside accounts departments.

Inkjets and lasers have moved on to become the preferred choice for almost all office departments. Inkjets offer superb colour resolution but do not print as sharp as a laser printer does—the output tends to bleed and smudge. Lasers, on the other hand, provide sharp printouts, but until not long ago, they simply lacked colour.

Companies therefore had inkjets for colour printing and lasers for black and white printing. There was one problem though—inkjets were proving to be a costly affair compared to laser printers. For perspective, consider that the accumulative cost incurred upon replacing the third successive ink cartridge surpasses the initial cost of the inkjet printer.

However, advertisers, design houses, and people in marketing departments had no option but to use inkjets for printing in colour.

Colour lasers were introduced a year ago in India, but they were simply too expensive as an option. As with many technologies, the price of lasers has come down to a level where they become a viable solution to small business

houses. Today, most colour lasers fall in the range of Rs 34,000 to Rs 50,000.

Further evolution has seen colour lasers shed their weight and lose their bulk in favour of a slimmer profile. The new breed of colour lasers can match mono lasers in speed, and are a little more expensive than their black and white counterparts. Older colour lasers took each document through four passes, one for each colour, which slowed them down. Now, a paper goes through just a single pass.

Though the new breed is classified as “laser” solutions, the underlying technology has changed, making them more compact and less expensive. Instead of a laser, these new printers have an array of LEDs that writes on the OPC—the organic photoconductive drum, the heart of a laser printer.

For this test we had colour lasers from the five top vendors: Canon, HP, Lexmark, Samsung and Wipro, with one model each. Xerox does have a model but we were not able to procure it on time for the test.

Along with these laser colour printers, we also tell you about some inkjet printers fresh out of the stables with unique features, strutting as business solutions. This mix of lasers and inkjets will give you a comprehensive look at available options for your business.



The Canon LBP-2410

How We Tested

Our test-bed comprised an Intel Pentium 4 3.2 GHz processor plugged on to an Intel 875PBZ motherboard, with 512 MB of Corsair 400 MHz DDR SDRAM, an MSI Ti4800 display card and a 40 GB, 7,200 RPM, ATA IV Seagate Barracuda hard drive.

We used Microsoft Windows XP with SP1 as the operating system, with the latest chipset and graphics drivers. The USB patch for Windows XP was installed, too, and the devices were connected on the USB 2.0 port for optimum performance. We used 100 gsm (gsm is grams per square metre—this measures the paper thickness) paper for the tests, except for the drain test, where we used 75 gsm papers. For photograph printing, we used NovaJet's 130 gsm inkjet photo paper across all the printers, since they are widely available across the country.

All the printers were tested on the

same test suite, which comprised a text document, a combi-document, a continuous document and an image file. The test files were chosen to reproduce every possible printing scenario.

The combi-document has many sections, with text of varying point sizes, lines drawn in a circle, seven coloured bars of decreasing colour gradients, and a small picture at one corner. Each of these subsections tests the printer's ability to reproduce detail. The varying text point size tests a printer's ability to reproduce the smallest possible text—the spatial resolution of the printer. The circle of lines tests a printer's ability to reproduce lines without jarring at the edges. Similarly, the seven-colour bar tests the colour reproduction capabilities of the printer's cartridges.

The continuous document is a

small report with bar graphs, pie charts, line graphs, images and text, all spread over six pages. This reproduces real-world usage.

The image document is made up of several elements with varying contrast and brightness areas. This document tests a printer's colour reproduction capability on photo paper.

The text and the combi-document file were printed at Normal and Best quality settings, and the image file was printed at the Best setting.

Scores granted to Features, Build Quality, Performance and Price were given weightages. An overall score out of 100 was then calculated. The product that scored the highest was adjudged the winner of the Digit Best Buy Gold award for its category. The runner-up was awarded the Digit Best Buy Silver award.